

Literature Review Summary

Team Number: 9

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Project Focus: How might we design an AI-enabled intuitive digital platform, including digital support groups or communities for first-time mothers, that provides easy-to-understand educational tools to learn and visualise each trimester of pregnancy, personalised health advice, pregnancy tracking, understanding of prenatal care requirements, and enables mothers to share experiences and seek peer support?

Overview

18 research papers were discussed regarding the design and development of our maternal health platform. The papers span across different methodological approaches, from strong conceptual frameworks to random controlled trials with several participants.

The geographic coverage concludes high-income contexts mostly UK and Germany coupled with low-resource settings like urban Indian slums, Dominican Republic, Caribbean immigrant communities, giving insights into both technological complexity and accessibility challenges.

Insights and Cross Cutting Patterns

A similar and strong pattern is being shown across all approaches talked about - the success of digital maternal health interventions depends much more on the thoughtful design than on technical complexity. For example the mMitra voice messaging service achieved 91% satisfaction among thousands of low-income Mumbai women not through advanced AI, but through culturally adapted content delivered in their local languages. In the same way the Baby Buddy app gained user trust because healthcare professionals appeared in their videos, making the platform feel more credible in spite of relatively having simple functionalities. The implications for our project are clear, advanced personal algorithms or predictive models add value only when it is layered on top of a culturally appropriate content. Chatbots demonstrated this failure mode when they generated responses which were at college reading levels, thus alienating the same populations which were in most need of support. Our platform must implement the progressive complexity whilst delivering the essential information in accessible formats first,

with sophisticated AI features enhancing rather than replacing human centered design. Digital support will mostly benefit the first-time mothers because of their higher knowledge gaps and anxiety. They prioritize accuracy, legibility, and most importantly practical utility over novelty, thereby rejecting the unreliable or other culturally irrelevant apps. The design of our platform's information architecture must be based upon this triad, we need to make sure that everything is medically accurate. The information also needs to be easy to grasp, which means we might have to simplify the reading level for several different users. To make this work well, the features must tackle the problems pregnant women directly face. That means focusing on the must have's functions and skipping anything which is extra.

Critical Divergences

Many research on digital health for maternal care tend to suffer from a "measurement gap": different communities like AI, mHealth, psychosocial, concentrate primarily on disparate, often proxy, measures such as accuracy, engagement, satisfaction, or symptom reduction, rather than strong clinical outcomes like mortality, complications, readmissions. This further prevents determining if the interventions improve health or just improve the patient experience. Our project must address this by tracking multi-level outcomes, to track clinical indicators alongside engagement metrics.

Limitations Across Literature

Current literature is limited by small, geographically specific, high-income western samples which tend to be of small sample sizes, thus lacking generalizability and only a few adequately powered studies have larger which is also around only thousands of patients. This hinders applicability in metrics like high-risk, low-resource settings with high maternal mortality. Our platform needs to fix this by getting much more data from groups we don't hear from often, which especially includes first-time mothers in tough situations. This will help us move past relying mostly on users who are currently urban, educated, and high-income. A lot of these intervention studies only run for a couple weeks, which is very short sighted view because pregnancy is mostly around 40 weeks long followed by postnatal period. Even though users are asking for longer involvement, this short term approach means we don't actually know if the behavioral changes actually stick or how useful the intervention is across journey of having a baby, hence our platform's retention must address these gaps by designing for longer months which cover conception to early postpartum, we will use features like progressive content, milestone celebrations, and evolving peer groups to support engagement and counteract any typical attrition.

Critical Research Gaps

Gaps exist in the integrated platforms which combine monitoring, communication, education, community support, and some personalized guidance, research in wearables, apps, chatbots, and peer support work in somewhat isolation, and interestingly no one's really looked into if these combined systems work better than just using the parts separately. Our platform must fill these gaps by integrating AI personalization, structured education, pregnancy tracking, and peer communities, we aim to test if this holistic approach achieves better outcomes than some fragmented tools, addressing the literature's suggestion that integration is most valuable and mostly mitigates user frustration from switching through multiple apps although empirical validation is currently lacking.

Mental health is mostly a peripheral and not a core part of prenatal care. Some papers discussed that pregnant women had depression signs, mediated by family functioning's role in stress. But monitoring platforms rarely track these psychological factors along with physical well-being, thus reflecting a divided approach to mind and body. The planned platform needs an integration between mental health screening and more support from the start. This could include regular talk, depression and anxiety assessments, dedicated peer emotional support spaces, and some AI content responsive to psychological as well as physical states.

Health literacy interventions increase knowledge but are rarely linked to changes in health behavior or reduced adverse outcomes. The link between these intermediate outcomes and maternal health is currently assumed and not proven *per se*. Our research must test these factors by tracking peer community engagement and clinical outcomes. This will then clarify if a specific peer interaction could be informational or emotional support, synchronous or asynchronous communication could be correlated with improved health, hence determining if peer support is a core clinical intervention or merely a social feature. Some of the existing maternal health technology design often stops just at prototyping without even measuring the actual health outcomes, thereby disconnecting user experience insights from the validated health improvements. Our method must integrate these insights and clinical outcome assessment in every design iteration to provide strong evidence on which the design patterns could truly improve maternal health and not just user satisfaction.

Implications for Our Platform Design

Effective maternal health platforms need to be culturally relevant which can be trusted for new mothers. Some of the key design elements are accuracy, legibility,

and practical utility throughout pregnancy. Most of the current systems fail to integrate the monitoring, education and personalization and peer support hence separating mental and physical health. The impact of peer support, health literacy, and design quality on clinical outcomes thereby remains unvalidated. Our platform should address these gaps thereby aiming to improve maternal health and strengthen the evidence base for digital tools.